Some controllability results for 1d linearized compressible Navier-Stokes Equations

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Compressible Navier-Stokes Equations

The Navier-Stokes equations for a compressible fluid in an interval $(0, L)$ $(L > 0)$ consists of:

$$\begin{cases} \partial_t \rho + \partial_x(\rho u) = 0, \\ \rho(\partial_t u + u\partial_x u) - \nu \partial_{xx} u - \partial_x P(\rho) = 0. \end{cases}$$

Here

- ρ is the density and u is the velocity
- $\nu > 0$ is the viscosity
- P denotes the pressure that satisfies (for isentropic/ isothermal fluid)

$$P(\rho) = a\rho^\gamma, \quad \text{for some } a > 0,\ \gamma \geq 1.$$

Control Problem

- We consider the following control problem:

$$(NS)\quad \begin{cases} \partial_t \rho + \partial_x(\rho u) = f\mathbb{1}_\omega & \text{in } (0,T)\times(0,L),\\ \rho(\partial_t u + u\partial_x u) - \nu\partial_{xx}u - \partial_x P(\rho) = g\mathbb{1}_{\mathcal{O}} & \text{in } (0,T)\times(0,L),\\ \rho(t,0) = p(t), \quad u(t,0) = 0, \quad u(t,L) = q(t), & \text{for } t\in(0,T),\\ \quad \text{or} & \\ \rho(t,0) = \rho(t,L) + p(t), & \\ u(t,0) = u(t,L) + q(t), \ \partial_x u(t,0) = \partial_x u(t,L), & \text{for } t\in(0,T),\\ \rho(0,x) = \rho_0(x), \quad u(0,x) = u_0(x), & \text{in } (0,L). \end{cases}$$

  where $T > 0$ and $\omega, \mathcal{O} \subset (0,L)$.

- We consider the trajectories $(\widetilde{\rho}, \widetilde{u})$ as solution to the system:

$$\begin{cases} \partial_t \widetilde{\rho} + \partial_x(\widetilde{\rho}\widetilde{u}) = 0,\\ \widetilde{\rho}(\partial_t \widetilde{u} + \widetilde{u}\partial_x \widetilde{u}) - \nu\partial_{xx}\widetilde{u} - a\gamma\widetilde{\rho}^{\gamma-1}\partial_x\widetilde{\rho} = 0. \end{cases}$$

Exact Controllability to the Trajectories

Exact controllability to the trajectories

Given $T > 0$ and (ρ_0, u_0), find control(s) $f, g, p, q_0, q_L, q_D, q_N$ such that the solution to (NS) satisfies

$$(\rho(T), u(T)) = (\widetilde{\rho}(T), \widetilde{u}(T)).$$

Local exact controllability to the trajectories

Given $T > 0$ and (ρ_0, u_0), find control(s) $f, g, p, q_0, q_L, q_D, q_N$ such that:
There exists a $\delta > 0$ such that if

$$\|(\rho_0, u_0) - (\widetilde{\rho}(0), \widetilde{u}(0))\| < \delta,$$

there exists a solution to (NS) satisfying

$$(\rho(T), u(T)) = (\widetilde{\rho}(T), \widetilde{u}(T)).$$

Controllability Results for NSE: Dirichlet Conditions

- Ervedoza, Glass, Guerrero and Puel [Arch. Rational Mech. Anal. (2012)] proved local exact controllability to constant states (ρ_*, u_*) with $\rho_* > 0, u_* \neq 0$ at time $T > \frac{L}{|u_*|}$ in the space $(H^3(0,L))^2$ using controls acting on the whole boundary.
- This result have been improved by Ervedoza and Savel [ESAIM: COCV (2018)] in the sense that
$$(\widetilde{\rho}, \widetilde{u}) \in [\mathcal{C}^2([0,T] \times [0,L])]^2 \text{ and } (\rho_0, u_0) \in (H^1(0,L))^2.$$
- In dimensions $N = 2, 3$, Ervedoza, Glass and Guerrero [Communications in PDEs, 2016] proved (for Ω smooth bounded domain) local exact controllability to the constant trajectory (ρ_*, u_*) with $\rho_* > 0, u_* \in \mathbb{R}^N \setminus \{0\}$ at time $T > \frac{L_0}{|u_*|}$ in the space $H^2(\Omega)^{1+N}$ using controls acting on the whole boundary, where
$$L_0 > \sup\left\{ \ell > 0 \ : \ \exists x \in \Omega \text{ s.t. } x + \ell \frac{u_*}{|u_*|} \in \Omega \right\}.$$

Controllability Results for NSE: In Torus

- Mitra, Ramaswamy and Renardy [Math. Meth. Appl. Sci. (2016)] proved local exact controllability to constant states (ρ_*, u_*) with $\rho_* > 0, u_* \neq 0$ at time $T > \frac{L}{|u_*|}$ in the space $(H^1_{\text{per}}(0, L))^2$ using a localized distributed control acting in the momentum equation.
- In dimensions $N = 2, 3$ for $\Omega \simeq \prod_{j=1}^{N}(0, L_j)$, Mitra and Renardy [Nonlinear Analysis: Real World Applications (2017)] proved local exact controllability to the constant state (ρ_*, u_*) with $\rho_* > 0, u_* \in \mathbb{R}^N \setminus \{0\}$ s.t. $(u_*)_1 \neq 0$ at time $T > \frac{L_1}{|(u_*)_1|}$ in the space $(H^2(\Omega))^{1+N}$ using a localized distributed control acting in the momentum equation.

Open Problems?

Exact controllability to the trajectories using

- a distributed control acting only in the continuity equation
- a control acting only on some part of the boundary (or, in 1d, using only one boundary control).

Still open! Difficulty in nonlinearities (hyperbolic behavior!)

First Step

Study the exact controllability to the trajectories of the linearized system

This is equivalent to the null controllability:

Null Controllability

Given $T, L > 0$ and (ρ_0, u_0), find control(s) $f, g, p, q_0, q_L, q_D, q_N$ such that the solution to (NS) satisfies

$$(\rho(T), u(T)) = (0, 0).$$

Problem Forumulation: Periodic Conditions

- Let $T > 0$. We consider the following linear control system around constant state (ρ_*, u_*) with $\rho_*, u_* > 0$:

$$(LNS) \quad \begin{cases} \partial_t \rho + u_* \partial_x \rho + \rho_* \partial_x u = f \mathbb{1}_\omega & \text{in } (0,T)\times(0,2\pi), \\ \partial_t u - \dfrac{\nu}{\rho_*}\partial_{xx} u + u_* \partial_x u + a\gamma\rho_*^{\gamma-2}\partial_x \rho = g\mathbb{1}_{\mathcal{O}} & \text{in } (0,T)\times(0,2\pi), \\ \rho(t,0) = \rho(t,2\pi) + p(t) & \text{for } t \in (0,T), \\ u(t,0) = \rho(t,2\pi) + q(t),\ \partial_x u(t,0) = \partial_x u(t,2\pi) & \text{for } t \in (0,T), \\ \rho(0,x) = \rho_0(x), \quad u(0,x) = u_0(x) & \text{in } (0,2\pi). \end{cases}$$

- $f, g \in L^2(0,T;L^2(0,2\pi))$ are localized distributed controls and $p, q_D \in L^2(0,T)$ denotes boundary controls.

Existing Results: Distributed Control

Distributed control acting only in the momentum equation:

- Chowdhury, Mitra, Ramaswamy and Renardy [J. Differential Equations, (2014)] proved:
  - There exists $g \in L^2(0,T;L^2(\mathcal{O})$ such that the system (LNS) is null controllable at time $T > \frac{2\pi}{u_*}$ in the space $\dot{H}^1_{\text{per}}(0,2\pi) \times L^2(0,2\pi)$.
  - If $0 \leq s < 1$, the system (LNS) is not null controllable at any $T > 0$ in the space $\dot{H}^s_{\text{per}}(0,2\pi) \times L^2(0,2\pi)$.

Distributed control acting only in the continuity equation:

- Beauchard, Koenig, Le Balc'h [J. Éc. polytech. Math, 2020] proved:
  - There exists $f \in L^2(0,T;L^2(\omega))$ such that the system (LNS) is null controllable at time $T > \frac{2\pi}{u_*}$ in the space $L^2(0,2\pi) \times \dot{L}^2(0,2\pi)$.
  - If $0 < T < \frac{2\pi}{u_*}$, the system (LNS) cannot be null controllable at time T in $(L^2(0,2\pi))^2$.

Existing Results: Boundary Control

Boundary control acting only on the velocity:

- Chowdhury and Mitra [J. Evol. Equ. (2015)] proved:
  - There exists $q \in L^2(0,T)$ such that the system (LNS) is null controllable at time $T > \frac{2\pi}{u_*}$ in the space $\dot{H}^{s+1}_{\text{per}}(0,2\pi) \times \dot{H}^{s}_{\text{per}}(0,2\pi)$ with $s > \frac{9}{2}$.
  - If $0 < T < \frac{2\pi}{u_*}$, the system (LNS) cannot be null controllable at time T in $(L^2(0,2\pi))^2$.

Question?

Null controllability of the system (LNS) at time $T > \frac{2\pi}{u_*}$ in the space $\dot{H}^{s+1}_{\text{per}}(0,2\pi) \times \dot{H}^{s}_{\text{per}}(0,2\pi)$ for $s \leq \frac{9}{2}$? (Not known!)

Boundary control acting only on the density: Not known!

New Results: Boundary Control

- Define the operator $(A, \mathcal{D}(A))$ as:

$$\begin{cases} A := \begin{pmatrix} -u_*\partial_x & -\rho_*\partial_x \\ -b\partial_x & \nu_0\partial_{xx} - u_*\partial_x, \end{pmatrix} \\ \mathcal{D}(A) := H^1_{\text{per}}(0, 2\pi) \times H^2_{\text{per}}(0, 2\pi), \end{cases}$$

where $\nu_0 := \frac{\nu}{\rho_*}$ and $b = a\gamma\rho_*^{\gamma-2}$.

Theorem (J.K.)

Let us assume that all the eigenvalues of A^* has geometric multiplicity 1. Then:

- **Control on density:**
  - *The system (LNS) is null controllable at time $T > \frac{2\pi}{u_*}$ in the space $(\dot{L}^2(0, 2\pi))^2$.*
  - *If $0 < T < \frac{2\pi}{u_*}$, the system (LNS) is not null controllable at any time $T > 0$ in $(L^2(0, 2\pi))^2$.*
- **Control on velocity:**
  - *The system (LNS) is null controllable at time $T > \frac{2\pi}{u_*}$ in the space $\dot{H}^1_{per}(0, 2\pi) \times L^2(0, 2\pi)$.*
  - *If $0 \leq s < 1$, the system (LNS) cannot be null controllable at any time $T > 0$ in $\dot{H}^s_{per}(0, 2\pi) \times L^2(0, 2\pi)$.*

Geometric multiplicity

- Null controllability at time $T > 0$ in $(\dot{L}^2(0, 2\pi))^2$ implies

$$\mathcal{B}^* \mathbf{\Phi}_\lambda := u_* \xi_\lambda(2\pi) + \rho_* \eta_\lambda(2\pi) \neq 0,$$

for all $\mathbf{\Phi}_\lambda := (\xi_\lambda, \eta_\lambda)$ eigenfunctions of A^* cor. to the eigenvalue λ.

- If $g.m.(\lambda) > 1$, then $\exists\ \widetilde{\mathbf{\Phi}}_\lambda, \widetilde{\widetilde{\mathbf{\Phi}}}_\lambda$ lin. ind. s.t. for

$$\mathbf{\Phi}_\lambda := C\widetilde{\mathbf{\Phi}} - D\widetilde{\widetilde{\mathbf{\Phi}}}, \text{ with } C := \mathcal{B}^* \widetilde{\widetilde{\mathbf{\Phi}}} \neq 0,\ D := \mathcal{B}^* \widetilde{\mathbf{\Phi}} \neq 0,$$

we have

$$\mathcal{B}^* \mathbf{\Phi}_\lambda = 0.$$

$\implies$ Not null controllable!

- If $\frac{4b\rho_*}{\nu_0^2} \notin \mathbb{Q}$ and $\frac{2\sqrt{b\rho_* - u_*^2}}{\nu_0} \notin \mathbb{N}$, then $g.m.(\lambda) = 1$ for all eigenvalue λ of A^*.
- If $\frac{2\sqrt{b\rho_* - u_*^2}}{\nu_0} \in \mathbb{N}$, there exists an eigenvalues λ of A^* such that $g.m.(\lambda) = 2$.

Problem Formulation: Dirichlet Conditions

- Let $T, L > 0$. We consider the following linear control system around constant state (ρ_*, u_*) with $\rho_*, u_* > 0$:

$$(LNS) \quad \begin{cases} \partial_t \rho + u_* \partial_x \rho + \rho_* \partial_x u = f \mathbb{1}_\omega & \text{in } (0,T)\times(0,L), \\ \partial_t u - \dfrac{\nu}{\rho_*}\partial_{xx} u + u_* \partial_x u + a\gamma \rho_*^{\gamma-2}\partial_x \rho = g\mathbb{1}_{\mathcal{O}} & \text{in } (0,T)\times(0,L), \\ \rho(t,0) = p(t), \quad u(t,0) = 0, \quad u(t,L) = q(t) & \text{for } t \in (0,T), \\ \rho(0,x) = \rho_0(x), \quad u(0,x) = u_0(x) & \text{in } (0,L). \end{cases}$$

- $f, g \in L^2(0,T; L^2(0,L))$ are distributed controls and $p, q \in L^2(0,T)$ denotes boundary controls.
- Transform: $\rho(t,x) \to \alpha\rho(\beta t, \delta x)$ and $u(t,x) \to u(\beta t, \delta x)$. Then: $(c = \frac{Q_0}{V_0}\sqrt{a\gamma Q_0^{\gamma-3}})$

$$\begin{cases} \partial_t \rho + \partial_x \rho + c\partial_x u = 0 & \text{in } (0,T)\times(0,\delta L), \\ \partial_t u - \partial_{xx} u + \partial_x u + c\partial_x \rho = 0 & \text{in } (0,T)\times(0,\delta L), \end{cases}$$

- W.l.o.g. we assume $\delta L = 1$.

New Results: Boundary Control on Density

Theorem (Bhandari, Chowdhury, Dutta, K.)

Let us assume that all the eigenvalues of A^* has geometric multiplicity 1. Then:

- *The system (LNS) is null controllable at time $T > 1$ in the space $\dot{L}^2(0,1) \times L^2(0,1)$.*
- *If $0 < T < 1$, the system (LNS) is not null controllable at any time $T > 0$ in $(L^2(0,1))^2$.*

- If $c^4 + 8c^2 + 5 < 4\pi^2$, then $g.m.(\lambda) = 1$ for all eigenvalue λ of A^*.

Questions?

- What happens if $c^4 + 8c^2 + 5 \geq 4\pi^2$? Does $\exists$ an eigenvalue λ of A^* s.t. $g.m.(\lambda) > 1$?

Future Directions and Open Problems

Dirichlet Conditions: Null controllability of the linearized system using

- a distributed control acting only in density/velocity equation
- a boundary control acting only on velocity

Local exact controllability to the trajectories using only one boundary control.

Thank you for your attention.

Happy Birthday! Professor Šárka.